

AlphaFold3 and AI Governance in Drug Candidate Selection

Case Study: AI Governance and Strategy

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Topic area: AlphaFold3: A neural network model using deep learning algorithms to determine 3-D structure of proteins to support expeditious drug candidate selection.

Summary of the case: AlphaFold3 is a deep learning algorithm that predicts 3-D structure of proteins. The previous version, AlphaFold2, predicted over 200 million protein structures, and the updated version, AlphaFold3, is currently used in drug development by supporting faster drug candidate selection. Traditionally, X-ray crystallography and cryo-electron microscopy have been used to determine protein structures; these methods are slower than AlphaFold3.¹ This approach uses amino acid sequence of proteins to predict their structures. 3-D structure of proteins provides an understanding of ligand binding sites to target those sites as potential drug candidates. Armed with such information, drug candidate selection and development are currently being approached.² Notably, Isomorphic Labs has partnered with drug companies to target small molecule development using this technology for undisclosed targets, according to press release in “BioSpace.” This deal is with Novartis and Eli Lilly drug companies. Due to the confidential nature of this deal, no other information can be found on drug candidates.³ Further potential for AlphaFold3 can be demonstrated by its extensive database on protein structures.⁴ As such, online journal search did find some potential drug candidates that have been identified using this tool, such as a novel discovery that targets CDK20, anti-cancer target.⁵ Another anti-cancer target compound is G490-0341 for development and research identified from protein structure found using AlphaFold, which is a WSB1 inhibitor.² We are years away from any drug being available for the general public using this technology because it takes about 10-15 years for the entire cycle of drug development to regulatory approval for human use.⁶ Still, AlphaFold3 would reduce the cost of drug development by building an effective target profile for drugs, once full development cycle is realized. Given the benefits of this technology, there are also downside to utilizing drug candidate selection with AlphaFold3 such as uncertainty of whether these drugs are as safe and effective as those selected by traditional methods, concerns of data privacy, and sponsor confidentiality issue should there be a need for public disclosure of their use of this technology.

Stakeholders:

Pharmaceutical industry: The drug industry is benefiting from AlphaFold3 protein fold predictions, as an ongoing effort. This targeted approach to drug candidate selection based on 3-D modeling of protein structures means that drug candidate selection can be done faster and more efficiently than traditional methods. There is a database of these structural predictions of proteins generated by AlphaFold3. This library is enabling the pharmaceutical industry to build drug candidates in an effective way armed with the knowledge of potential ligand/receptor interactions of these structures. Such collaborative efforts means that the cost of drug production is being reduced along with decreased time to produce an effective compound, given benefits potentially being passed to the general consumer.

Health Authorities: There is no consensus among health authorities on how AlphaFold3 should be used, or a concrete regulatory framework applicable across the board. The FDA does have a draft guidance document on AI use to support regulatory decisions for drugs/biologics.⁷ The Agency's current thinking doesn't cover how AlphaFold3 should be utilized. Technical or ethical challenges with its use is not addressed either. Health authorities would need to develop these standards, as the use of this AI tool becomes commonplace for drug candidate selection, to safeguard the public for any potential issues through further guidance documents to enhance existing regulations.

ICH: To standardize the use of tools like AlphaFold3, ICH would need to develop guidance documents that would be utilized across the board for drug development. Generally, ICH guidance documents are required to be followed by the drug industry, even if it is not codified. As such, the guidance would need to address concerns of GxPs, to support regulatory, ethical, technical, and safety matters of AlphaFold3 for drug candidate selections globally.

Note: ICH is not a health authority or a regulatory body, like the FDA. It is a committee, which provides guidance to the pharmaceutical industry globally. Health authorities are regulatory bodies. The ICH guidance documents are different than those generated by a health authority, such as the FDA's guidance documents. ICH is global, and the FDA is for US. Other health authorities are also country specific, like the US FDA.

Connection to broader issues:

Use of AlphaFold3 for drug candidate selection comes with broader issues in technology and ethical concerns. There is no guarantee that selected target sites would be effective and safe. Drugs undergo clinical trials with human subjects, to ensure safety and efficacy before approval. Benefits/risks are not fully realized until real-world data is available, after approval. As such, this tool has the potential to select drug candidates that are less safe and effective, compared to traditional means of drug candidate selection. Traditionally, drug candidates are screened using techniques to find those that qualify for acceptable safety, efficacy, and pharmacokinetic properties.⁸ Human subjects are then exposed to unsafe drugs during testing phase if selected drug favors risk, and this would be seen after market approval of the drug, when enough real-world data is collected to realize this to pull the drug off of the market. Given this novel technology, the unblinding of patients' personal/medical information risk is there, should there be a need due to protect them from harm due to the unsafe product generated using this tool. This poses an issue with data privacy because this unblinding activity has not caught up with regulations by health authority(ies). It leads to legal issues for sponsors of the drug made using this AI tool as well. Companies developing drugs with this tool are not disclosing what drug targets they are developing, so this poses issue with sponsor confidentiality. Does the public have the right to know which drugs in the market or in research is made using this tool to understand the technical and ethical challenges of this technology. Another case study that can connect to

these broader issues is using AI to support patient enrollment in clinical trials. PPD is using AI for enrolling patients in clinical trials to speed up drug development process.⁹ Potential for unsafe drug use, data privacy, and sponsor confidentiality concerns is seen with clinical trial recruitment support with AI as well.¹⁰

Importance:

This case deserves attention, and it is important to resolve ethical issues surrounding the technology use of AlphaFold3. The most regulated industry in the world, including the US, is the drug industry. The reason is that drugs are generally made for human use, so they need to be safe and effective. It is unclear drugs selected using this tool are all safe and effective. This tool cannot predict point mutations in proteins or post-translational modifications of proteins. It also cannot fully understand molecules that interact with proteins.¹¹ As such, drug targets can have issues, but this does not negate the fact that there are many diseases that need drug targets that are currently unknown. Given this, AlphaFold3 can support accelerated drug candidate selection even with its flaws to support cost reduction in drug development to speedy early phase of drug research. By fully understanding 3-D structures of proteins using this tool, a clear picture of drug binding sites for proteins is revealed to support drug targets.¹² Use of this tool would mean that diseases that have unknown target sites can benefit with drug candidate selection, given the database of AlphaFold3 for protein structures. Concerns of this technology and ethics are not adequate to limit its use. They can be addressed with policies/regulations by ICH and health authorities.

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Appendix:

1. Health authority(ies): could mean the US FDA or any other country version of this regulatory body.
2. ICH: The International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use. (Harmonisation: is the correct spelling.)
3. Biologics: drugs made using living organisms. Considered large molecules.
4. Drugs: compounds made using synthetic means. Considered small molecules.
5. Agency: In this context, it is referring to the US FDA.
6. AlphaFold3: Current version of AlphaFold. It has been updated from AlphaFold2, to AlphaFold3.
7. Guidance documents: The US FDA has their own guidance documents, that is considered their current thinking, and will need to be followed, unless an alternative that is equal or better exist by the drug industry. The Agency's guidance documents can be in draft (comment period) or in final state; either case, they would need to be adhered to, since they are technically binding by the Agency's standards. Additionally, ICH also has their own

guidance documents, that have global impact in the drug industry. They too need to be followed.

8. GxPs: three main GxPs are GLP (Good Laboratory Practice), GMP (Good Manufacturing Practice), and GCP (Good Clinical Practice). They work together to ensure that drugs produced are safe and effective and follow regulations for human consumption. GxPs are adhered globally by the drug industry as a requirement for life cycle of drug development.
9. Codified: This generally means it is in regulations, such as 21CFR for the FDA.
10. 21CFR: FDA regulations. CFR: Code of Federal Regulations.
11. Small molecule: drugs, not biologics.
12. Sponsors: drug companies.
13. Human subjects: those enrolled in clinical trials.
14. PPD: one of the leading CROs (CRO: Contract Research Organization (contracted by pharmaceutical companies to conduct clinical trials.))
15. Patients: those in clinical trials.